



Root finding Problems

- Root-Finding Problem is the problem of finding a root of the equation $f(x) = 0$, where $f(x)$ is a function of a single variable x .
- **Given a function $f(x)$, find $x = r$ such that $f(r) = 0$.**
- The equation, will be called algebraic or transcendental according as $f(x)$ is purely a polynomial in x or contains some other functions, such as trigonometric, logarithmic or exponential functions etc. For example

$$x^3 - x + 2 = 0$$

algebraic equation

$$\sin(x) + x^2 - \log x = 0$$

transcendental equation

Roots of Equations/Zeros of a function

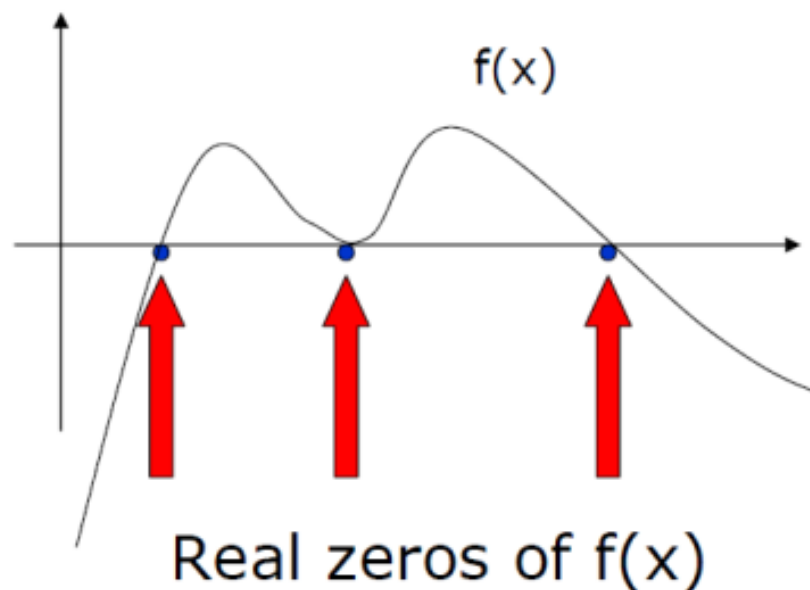
- A number r that satisfies an equation is called a root of the equation.

The equation $x^4 - 3x^3 - 7x^2 + 15x = -18$
has four roots $-2, 3, 3$ and -1

- Let $f(x)$ be a real-valued function of a real variable. Any number r for which $f(r)=0$ is called a zero of the function.
- Examples: 2 and 3 are zeros of the function
 $f(x) = (x-2)(x-3)$

Graphical Interpretation of zeros

- The real zeros of a function $f(x)$ are the values of x at which the graph of the function crosses (or touches) the x -axis.



Solution Methods

Several ways to solve nonlinear equations are possible.

- **Analytical Solutions:** possible for special equations only
- **Graphical Solutions:** Useful for providing initial guesses for other methods
- **Numerical Solutions:**
 - ❖ Bisection Method
 - ❖ Newton's Method
 - ❖ Secant Method
 - ❖ False position Method
 - ❖ Fixed point iterations

Basis of Bisection Method

- **Theorem:** An equation $f(x) = 0$, where $f(x)$ is a real continuous function, has at least one root between a and b if $f(a)f(b) < 0$.

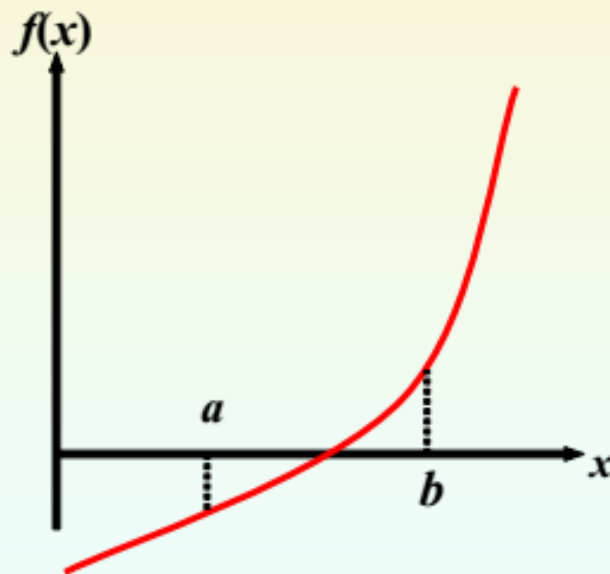


Figure 1 At least one root exists between the two points if the function is real, continuous, and changes sign.

Basis of Bisection Method

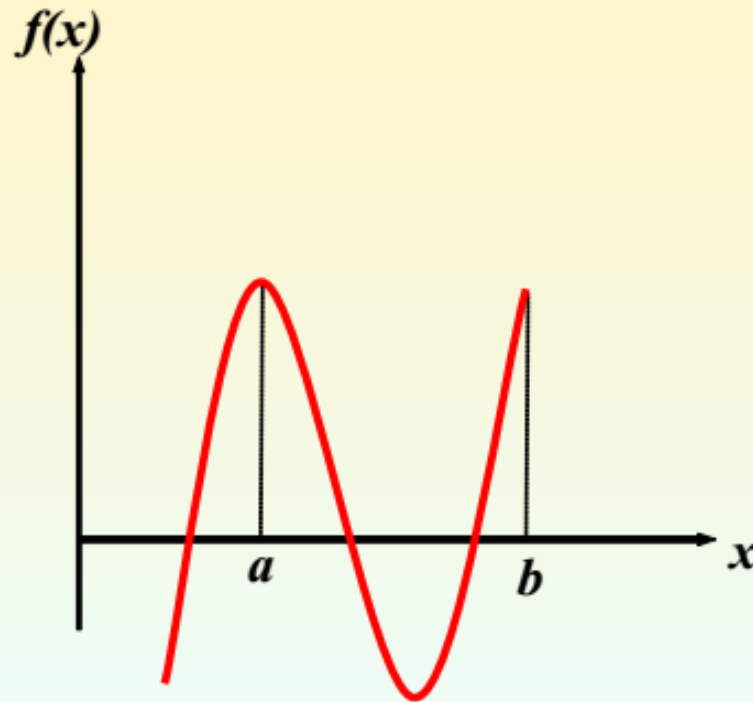


Figure 2 If function does not change sign between two points, roots of the equation may still exist between the two points.

Basis of Bisection Method

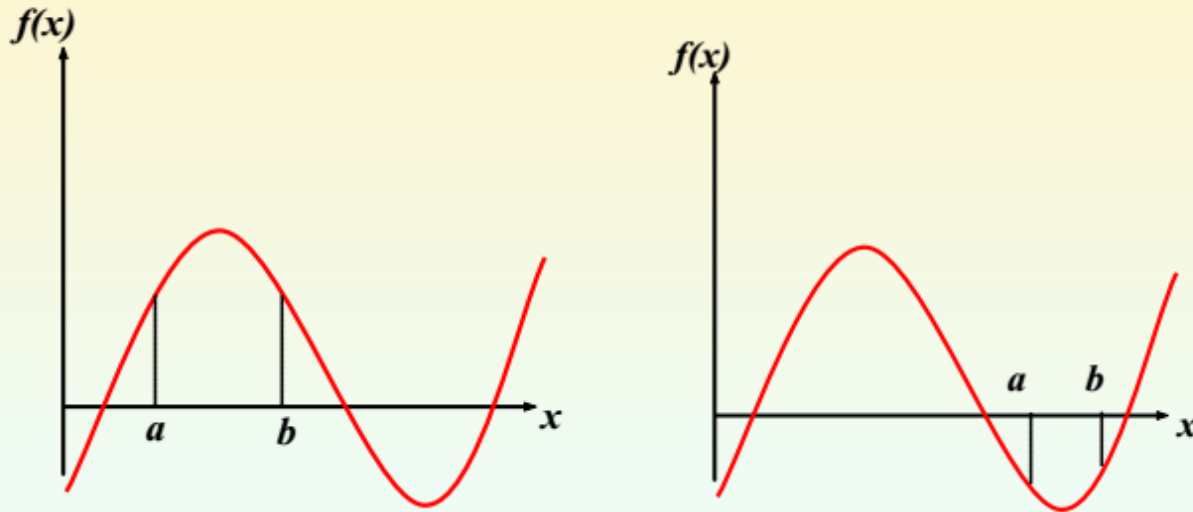


Figure 3 If the function does not change sign between two points, there may not be any roots for the equation between the two points.

Basis of Bisection Method

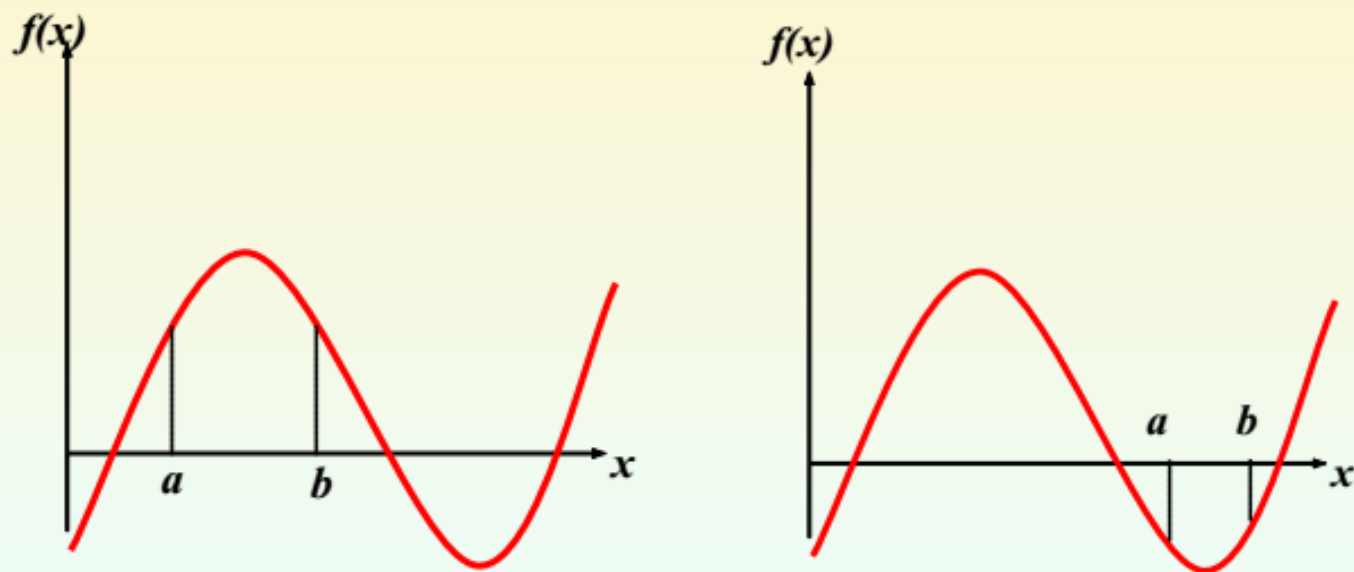


Figure 3 If the function $f(x)$ does not change sign between two points, there may not be any roots for the equation $f(x)=0$ between the two points.

Basis of Bisection Method

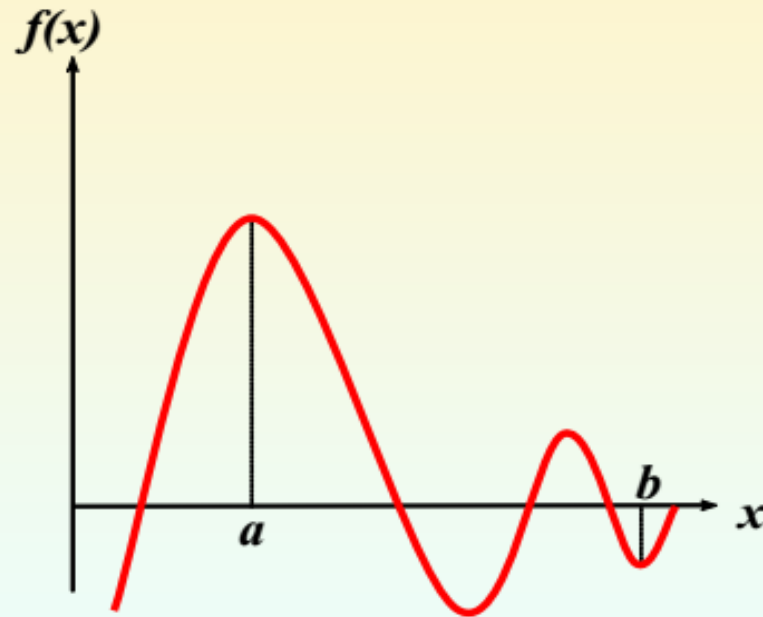
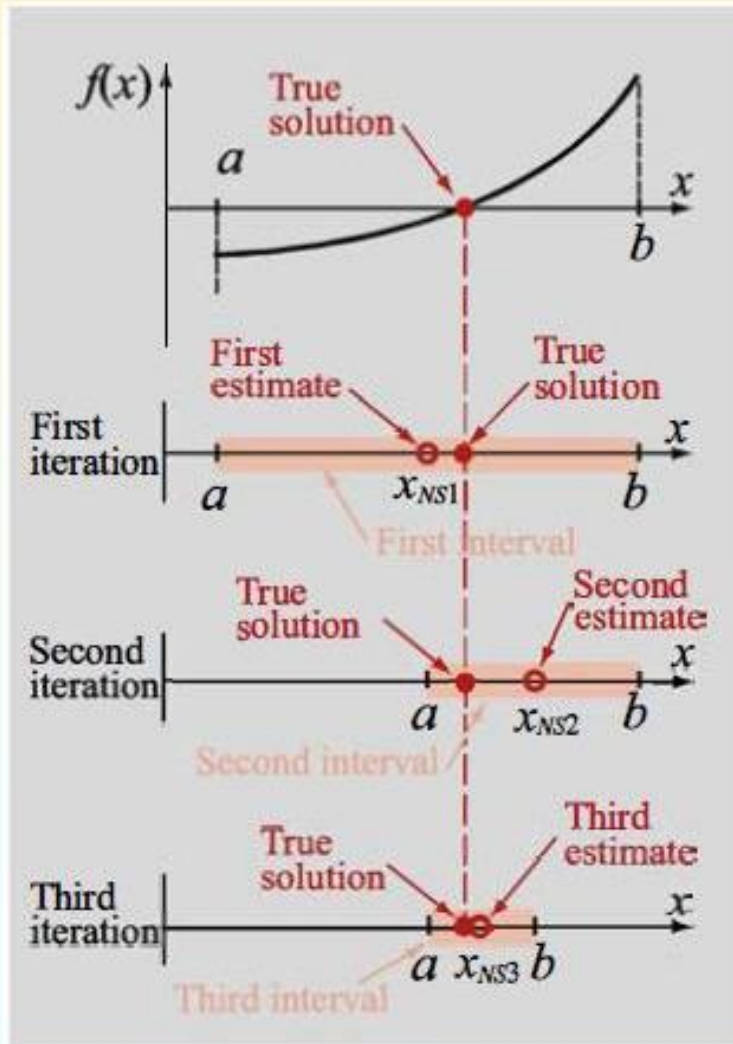


Figure 4 If the function $f(x)$ changes sign between two points, more than one root for the equation $f(x)=0$ may exist between the two points.

Basis of Bisection Method



Basis of Bisection Method

- **The Bisection method is one of the simplest methods to find a zero of a nonlinear function.**
- **It is also called interval halving method.**
- **To use the Bisection method, one needs an initial interval that is known to contain a zero of the function.**
- **The method systematically reduces the interval. It does this by dividing the interval into two equal parts, performs a simple test and based on the result of the test half of the interval is thrown away.**
- **The procedure is repeated until the desired interval size is obtained.**

Method of Bisection

- If the function is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have different signs, Bisection Method obtains a new interval that is half of the current interval and the sign of the function at the end points of the interval are different.
- This allows us to repeat the Bisection procedure to further reduce the size of the interval.

Algorithm of Bisection Method

- **Inputs:** (i) $f(x)$ - The given function
(ii) $a; b$ - The two numbers such that $f(a)f(b) < 0$
- **Output:** An approximation of the root of $f(x) = 0$ in $[a; b]$
- **For** $k = 0; 1; 2; : : :$, **do until** satisfied:
 Compute $c_k = (a_k + b_k)/2$.
 Test if c_k is the desired root, if so, stop.
 If c_k is not the desired root, test if $f(c_k)f(a_k) < 0$
 If so, set $b_{k+1} = c_k$ and $a_{k+1} = a_k$.
 Otherwise, set $a_{k+1} = c_k$; $b_{k+1} = b_k$.
- **End**

Step 1

Choose a and b as two guesses for the root such that $f(a)f(b) < 0$, or in other words, $f(x)$ changes sign between a and b . This was demonstrated in Figure 1.

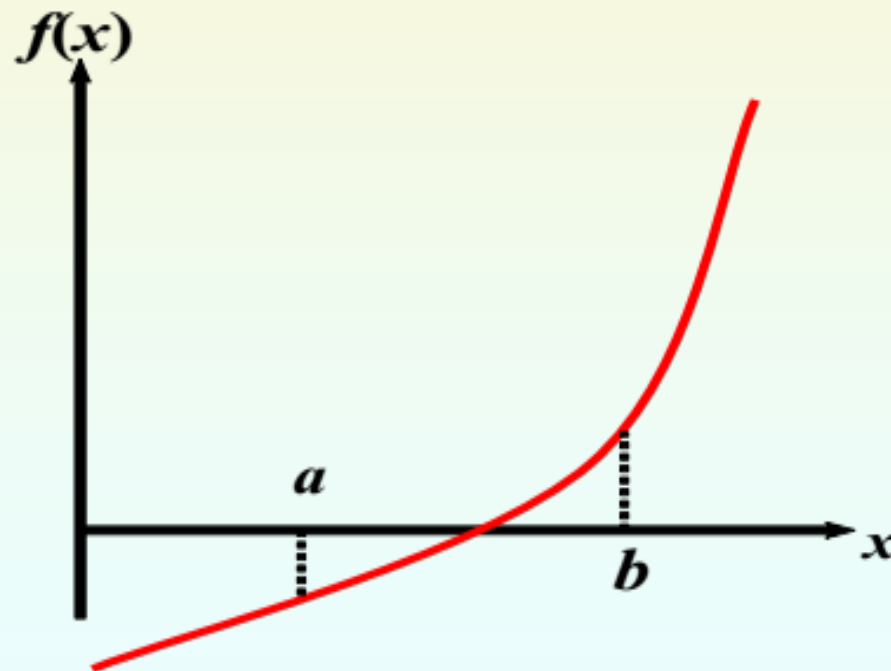


Figure 1

Step 2

Estimate the root, c of the equation $f(x) = 0$ as the mid point between a_k and b_k as

$$c_k = \frac{a_k + b_k}{2}$$

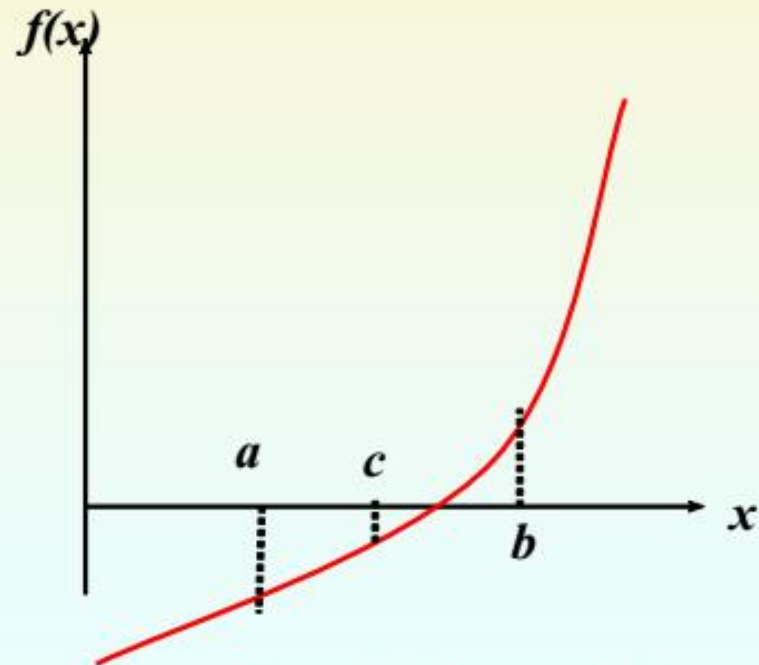


Figure 5 Estimate c

Step 3

Now check the following

- a) If $f(a_k)f(c_k) < 0$, then the root lies between a_k and c_k ; then $a_{k+1} = a_k$; $b_{k+1} = c_k$.
- b) If $f(a_k)f(c_k) > 0$, then the root lies between c_k and b_k ; then $a_{k+1} = c_k$; $b_{k+1} = b_k$.
- c) If $f(a_k)f(c_k) = 0$; then the root is c_k . Stop the algorithm. c_k is the desired root.

Step 4

Find the new estimate of the root

$$c_{k+1} = \frac{a_k + b_k}{2}$$

Find the absolute relative approximate error

$$|\epsilon_a| = \left| \frac{c_{k+1} - c_k}{c_{k+1}} \right| \times 100$$

where

c_k = previous estimate of root

c_{k+1} = current estimate of root

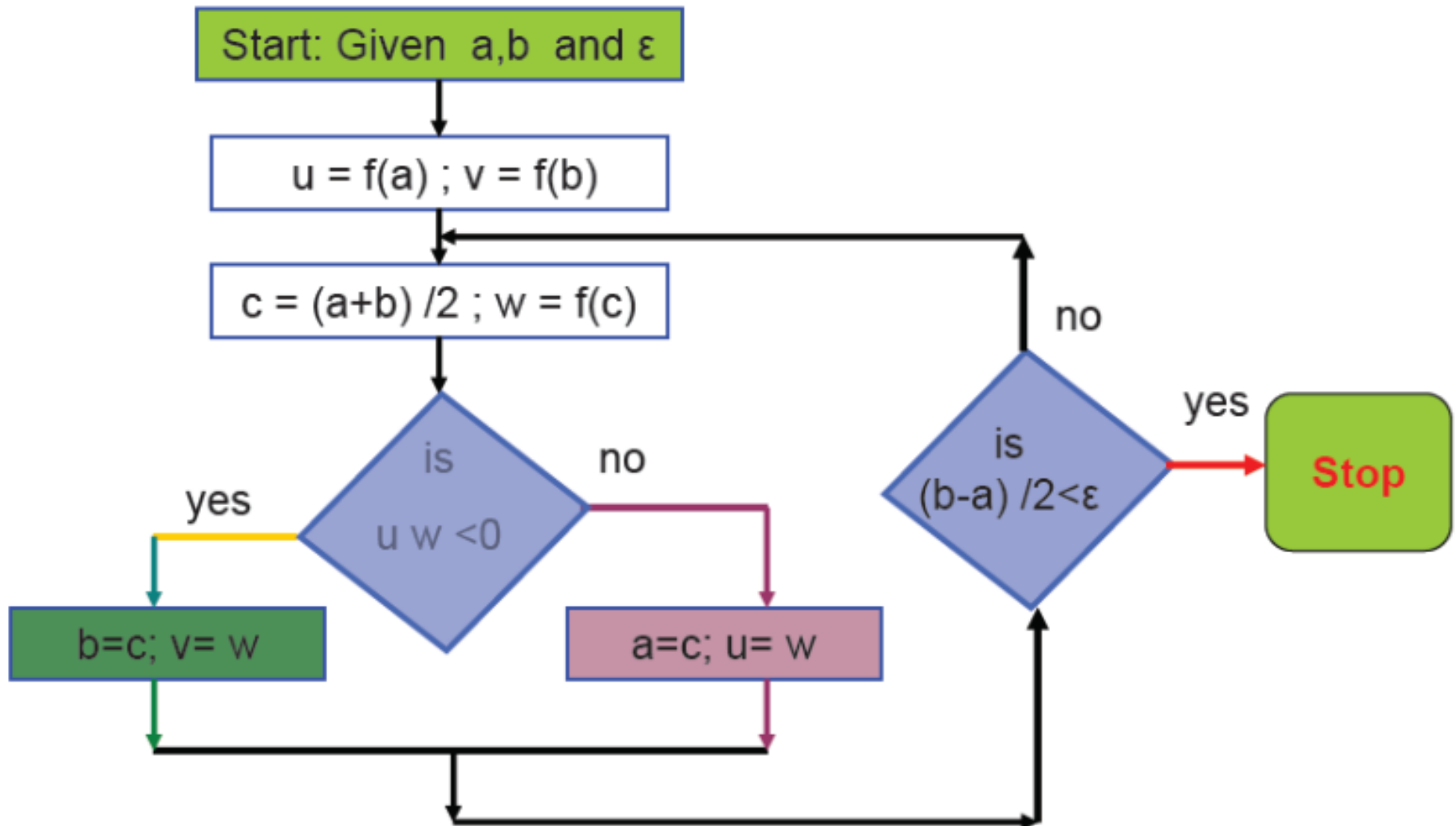
Stopping Criteria

- Since this is an iterative method, we must determine some stopping criteria that will allow the iteration to stop. Here are some commonly used stopping criteria. Let ε be the tolerance; that is, we would like to obtain the root with an error of at most of ε . Then

Accept $x = c_k$ as a root of $f(x) = 0$ if any of the following criteria is satisfied:

1. $|f(c_k)| \leq \epsilon$ (*The functional value is less than or equal to the tolerance*).
2. $\frac{|c_{k-1} - c_k|}{|c_k|} \leq \epsilon$ (*The relative change is less than or equal to the tolerance*).
3. $\frac{(b-a)}{2^k} \leq \epsilon$ (*The length of the interval after k iterations is less than or equal to the tolerance*).
4. *The number of iterations k is greater than or equal to a predetermined number, say N .*

Flow chart of Bisection Method



Example: The Bisection Method

Show that $f(x) = x^3 + 4x^2 - 10 = 0$ has a root in $[1, 2]$ and use the Bisection method to determine an approximation to the root that is accurate to at least within 10^{-4} .

Solution

- Because $f(1) = -5$ and $f(2) = 14$ the Intermediate Value Theorem ensures that this continuous function has a root in $[1, 2]$.
▶ IVT
- For the first iteration of the Bisection method we use the fact that at the midpoint of $[1, 2]$ we have $f(1.5) = 2.375 > 0$.
- This indicates that we should select the interval $[1, 1.5]$ for our second iteration.
- Then we find that $f(1.25) = -1.796875$ so our new interval becomes $[1.25, 1.5]$, whose midpoint is 1.375.
- Continuing in this manner gives the values shown in the following table.

Example: The Bisection Method

Show that $f(x) = x^3 + 4x^2 - 10 = 0$ has a root in $[1, 2]$ and use the Bisection method to determine an approximation to the root that is accurate to at least within 10^{-4} .

Iter	a_n	b_n	p_n	$f(a_n)$	$f(p_n)$	RelErr
1	1.000000	2.000000	1.500000	-5.000	2.375	0.33333
2	1.000000	1.500000	1.250000	-5.000	-1.797	0.20000
3	1.250000	1.500000	1.375000	-1.797	0.162	0.09091
4	1.250000	1.375000	1.312500	-1.797	-0.848	0.04762
5	1.312500	1.375000	1.343750	-0.848	-0.351	0.02326
6	1.343750	1.375000	1.359375	-0.351	-0.096	0.01149
7	1.359375	1.375000	1.367188	-0.096	0.032	0.00571
8	1.359375	1.367188	1.363281	-0.096	-0.032	0.00287
9	1.363281	1.367188	1.365234	-0.032	0.000	0.00143
10	1.363281	1.365234	1.364258	-0.032	-0.016	0.00072
11	1.364258	1.365234	1.364746	-0.016	-0.008	0.00036
12	1.364746	1.365234	1.364990	-0.008	-0.004	0.00018
13	1.364990	1.365234	1.365112	-0.004	-0.002	0.00009

Theorem

Suppose that $f \in C[a, b]$ and $f(a) \cdot f(b) < 0$. The Bisection method generates a sequence $\{p_n\}_{n=1}^{\infty}$ approximating a zero p of f with

$$|p_n - p| \leq \frac{b - a}{2^n}, \quad \text{when } n \geq 1.$$

Examples

- Find a root of $f(x) = x^3 + x^2 - 1 = 0$ correct to 4-decimal places
- Find a root of $f(x) = \cos -1.3x = 0$ correct to 3-decimal places
- Find a root of $f(x) = e^x - x^2 + 3x - 2 = 0$ correct to 3-decimal places
- Find a root of $f(x) = 3x + \sin x - e^x$ correct to 4-decimal places
- Find a root of $f(x) = x^4 - 2x^3 - 4x^2 + 4x + 4 = 0$ correct to 4-decimal places